MODEL SENSE · ALM MODEL LAB

Model Sense

ALM Model Lab — User Guide

A browser-based teaching lab for interest-rate simulation, behavioral instrument modeling, and asset-liability analytics. This edition pairs the operating manual with the concepts, functional forms, and parameter defaults of every model.

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CONTENTS

Orientation	1
Notation and conventions	2
Market Data	3
Rate Models	5
Instruments	6
ALM Analytics	12
Glossary	16

Orientation

What the Lab is, how the four sections fit together, the order to work through them, and how to read the technical material.

What the Lab does

The ALM Model Lab takes a market snapshot, calibrates two interest-rate models to it, simulates forward rate paths, and prices a small book of bank instruments and behavioral deposits against those paths. Everything runs in your browser; nothing is uploaded.

It is a teaching tool. The numbers are illustrative and the defaults are pedagogical, not a production risk system. The point is to make the mechanics of rate simulation, prepayment, deposit behavior, replicating portfolios, and funds transfer pricing visible and manipulable.

The four sections, in working order

The top navigation has four sections. Work left to right the first time through.

- **Market Data.** Import a snapshot, review the rate history, bootstrap the benchmark curve, and inspect the volatility surface (SABR). Calibration of the rate models happens here.
- **Rate Models.** Run the Hull-White and BGM simulations and compare them, then the two behavioral transmission models that map benchmark rates into administered rates: the SFR client rate and the MMA deposit beta. The market engines produce the path sets every downstream analytic consumes.
- **Instruments.** Parameterize the product cash flows: the fixed loan, floating loan, mortgage, the two non-maturity deposit books, and the CD book.
- **ALM Analytics.** Repricing gap, liquidity gap, the deposit replicating portfolio, funds transfer pricing (loans and NMD), effective duration, and the stochastic earnings gap. This is where the book-level results live.

The typical workflow

A full pass is: import a snapshot on the Import tab, open the SABR tab and click Calibrate models, then run the Hull-White and BGM simulations on their tabs. With both simulations in hand, every Instruments and Analytics tab is live. The FTP and Replicating Portfolio tabs need both simulations because they optimize and option-price over the simulated paths.

The header shows the loaded calibration date and path count. The info button (top right) opens a popover with the calibrated model parameters and known limitations.

How to read the technical material

Each section below pairs the operating instructions with the functional form and the parameter defaults of the models that drive it. Read the Notation and conventions section first: it fixes the units, the sign convention, the rate-path bridge, and the two shock conventions that the rest of the guide assumes throughout.

Formulas are written exactly as the engine computes them. Where the Lab simplifies a canonical method for tractability or teaching, the simplification is stated rather than hidden. Defaults are the values shipped on the tabs; almost all are editable.

Notation and conventions

Units, signs, the rate-path bridge, the Monte Carlo defaults, and the EVE-versus-NII shock conventions that every model below assumes.

Units and sign conventions

The Lab mixes annualized rates with monthly behavioral intensities by design, so units are load-bearing. Read a formula by first checking what each quantity is denominated in.

- **Curve rates.** Decimal per annum (0.04 = 4%). Zero rates are continuously compounded; discount factors are $P(t) = \exp(-z(t)\cdot t)$.
- **Deposit and decay intensities.** Closure and rate-driven decay are in percent per month and are divided by 100 before they compose; deposit rates and spreads are in percent per annum.
- **Time grid.** Monthly. A cashflow at month m sits at t = m/12 years. The fixed-leg day-count is ACT/360, applied as 365/360 per full year.
- **Balances.** Behavioral runoff fans are normalized to a start of 100; cashflow and PV tables are in dollars.
- **Sign.** Assets carry a positive sign and liabilities negative in the gap and SEG views. FTP margins are constructed to be positive franchise value on both sides: asset NIM is yield minus all-in FTP, liability credit is all-in FTP minus deposit cost.

The rate-path bridge

Every instrument is priced against a RatePath: an object exposing a short rate at each monthly step and a term forward at any tenor. A deterministic path reads the central bootstrapped curve; a stochastic path wraps a single Hull-White or BGM realization. The same generator code runs in both, so the deterministic line and the Monte Carlo mean are produced identically.

The one subtlety a practitioner must hold: term forwards beyond one year are read as par-swap rates, not as simple-compounded money-market forwards, because simple compounding overstates multi-year rates by hundreds of basis points. The split happens at one year.

> $$f(t1, t2) = (P(t1)/P(t2) - 1) / (t2 - t1)$$
> simple-compounded forward, money-market convention; used only for tenors up to 1Y
>
> $$S(t1, t2) = (P(t1) - P(t1+n)) / \Sigma\_k\ (365/360)\cdot P(t1+k), \quad n = round(t2 - t1)$$
> forward par-swap rate, annual ACT/360 fixed leg; used for tenors beyond 1Y

Monte Carlo defaults and the martingale correction

Both simulators run 500 paths as 250 antithetic pairs over a 30-year monthly horizon from a fixed seed, so a given snapshot reproduces bit-for-bit. Antithetic pairing draws one standard normal per pair and applies it with both signs, halving the variance of any quantity that is roughly linear in the shock.

After simulation, a multiplicative correction rescales the per-step discount factors so the simulated mean discount factor equals the market discount factor at every horizon. This removes the small numerical drift of the discretized scheme; it is distinct from the analytic convexity term inside the bond formula. The raw, pre-correction error is reported as a diagnostic.

> $$correction\_k = P\hat{\ }M(0, t\_k) / E\_p[\ \tilde{D}\_p(t\_k)\ ]$$
> P^M is the market discount factor; E_p the path-mean of the uncorrected simulated discount factor at step k

Two shock conventions: EVE versus NII

A parallel rate shock means two different things depending on what you are measuring, and the Lab keeps them separate on purpose.

For an economic-value (EVE) or effective-duration view, the shock applies from step 0: the entire rate environment moves instantly, including the first coupon. For an earnings (SEG / NII) view, the first month's coupon is locked before the shock event, so step 0 is left unshocked. The discount-side shock is a parallel shift of the zero curve; the cashflow-side shock re-simulates the behavioral response, so prepayment and deposit beta react to the new path rather than merely being re-discounted.

Market Data

Import, rate history, curve bootstrap, the term liquidity premium, and the SABR volatility surface.

Import

Loads a market snapshot: the benchmark curve quotes, cap and swaption volatilities, the term funding curve, and the term liquidity premium nodes. A default snapshot loads automatically so you can explore without a file. You can drop your own workbook to override it.

Cap and swaption volatilities are normal (Bachelier, basis-point) vols, not lognormal. Cap strikes are absolute rates and may be negative, reflecting the negative-rate-era convention; the at-the-money cap is carried as its own column rather than inferred from a zero strike.

Rate History and the deposit-tractor pillar

The historical benchmark term series drives the replicating-portfolio moving averages. The steady-state yield of a k-month pillar is the trailing k-month average of the k-month tenor, so the history is what gives the deposit tractor its carry lag.

Two details matter when reconciling against the replicating portfolio. The pillar moving average is inclusive of the as-of month, which differs from the strictly-lagged window the Non-IB deposit model uses for its regime moving average; do not conflate the two. And the shipped history is published data, not a reconstruction: month-end US Treasury CMT par yields from the FRED DGS series, 2008-10 through 2026-03. The short anchor is the 1-month bill CMT, and the term funding curve is that same CMT curve plus a disclosed, hand-specified spread schedule rather than a modeled SOFR term structure.

> $trailingMA(values, i, k) = (1/k) \cdot \Sigma_{j = i-k+1 .. i}\ values[j]$
>
> inclusive trailing k-month average at index i; NaN before the first complete window
>
> $pillarYield(asOfIdx, k) = trailingMA(\ sofrTermSeries(k), asOfIdx, k\ )$
>
> steady-state yield of the k-month tractor pillar, decimal per annum

Curve

Bootstraps the benchmark zero curve from the snapshot quotes. Short-end nodes at or below one year invert directly from a simple-compounded discount factor; longer nodes are solved one at a time so each maturity reprices to par given the shorter nodes already fixed. The solve is an exact root-find per node, not a least-squares fit, and interpolation is linear in the zero rate, which makes a parallel shock exact at every node.

Every downstream analytic (FTP, EVE, cap pricing, the simulation drivers) reads this one curve, so there is a single object of truth. The float leg uses the textbook par identity rather than projecting index fixings, the standard single-curve OIS simplification where the discount and forward curves coincide.

> $df = 1 / (1 + r \cdot t),\quad z = -ln(df) / t$
>
> short-end inversion, tenor t up to 1Y; r is the quoted simple rate, z the continuously-compounded zero
>
> $1 - exp(-z\_N \cdot t\_N) = \Sigma\_k\ r \cdot (365/360) \cdot exp(-z(c\_k) \cdot c\_k)$
>
> par-swap node equation for tenors beyond 1Y; coupon dates c_k = 1..n with n = round(t_N); solve z_N by Brent root-find
>
> $shockCurve: z(t) \to z(t) + shockBp/1e4$
>
> the EVE discount-side shock; a parallel shift of every node, so discount factors scale by exp(-s·t)

Term liquidity premium and the all-in funding curve

The term liquidity premium (TLP) is the spread of the institution's wholesale term-funding curve over the matching-tenor benchmark: the term funding cost a desk actually pays (for a U.S. bank the classic example is FHLB advance rates). It is pinned to zero overnight, since the bank funds overnight at policy rates rather than through term borrowing, and rises with tenor. It composes additively with the benchmark zero curve to form the all-in funding curve that FTP charges and credits against, which cleanly separates the pure-rate leg from the liquidity leg.

The shipped node set is the constructed illustrative spread schedule from the default snapshot (hand-specified round values, rising with tenor), interpolated linearly and held flat outside the bracket. Import your own market workbook to replace it with your institution's actual advance-curve spreads.

> $z\_all\text{-}in(t) = z\_sofr(t) + tlp(t)$
>
> all-in funding curve; TLP applied to discounting as exp(-tlp(t)·t)

- **Node set.** Overnight 0, 1M 21 bp, 3M 19 bp, 6M 20 bp, 1Y 31 bp, 2Y 43 bp, 3Y 50 bp, 5Y 56 bp, 10Y 91 bp, 30Y 153 bp. Linear in spread, clamped flat at the ends.

SABR volatility surface

The SABR tab shows the normal-vol smile and triggers the two model calibrations. The smile is the Hagan 2002 normal-vol expansion in its beta-equals-zero (Bachelier) limit, evaluated from a parameter set chosen by a strike-coverage heuristic rather than fit by least squares, so treat the displayed surface as a smile interpolant, not a market-fitted SABR. The genuine least-squares fits are Hull-White to the at-the-money cap column and BGM to the swaption surface, both run when you click Calibrate models. The rate-model tabs stay disabled until that calibration has run.

$$\sigma\_N(ATM) = \alpha \cdot \left(1 + ((2 - 3\rho^2)\cdot\nu^2/24)\cdot T \right)$$

at-the-money normal vol; α is the level (vol of rate), ν the vol-of-vol, ρ the skew, T the expiry

$$z = (\nu/\alpha)\cdot(F - K), \quad x = \ln\left((\sqrt{(1 - 2\rho z + z^2)} + z - \rho) / (1 - \rho) \right)$$

away-from-ATM: F is the forward, K the strike

$$\sigma\_N = \alpha \cdot (z/x) \cdot \left(1 + ((2 - 3\rho^2)\cdot\nu^2/24)\cdot T \right)$$

the ATM branch is the z → 0 limit where z/x → 1

Rate Models

Hull-White and BGM simulation, their calibration and dynamics, and the engine comparison.

Hull-White (1-factor)

An exact-transition short-rate simulation. The short rate is the bootstrapped instantaneous forward plus a mean-reverting Gaussian factor, so the model is curve-consistent at every horizon by construction. The factor follows an Ornstein-Uhlenbeck process that is sampled by its exact one-step transition, so there is no time-discretization bias in the factor itself; the only discretization is the trapezoidal accrual used for discounting, which the martingale correction then removes.

Hull-White projects any tenor analytically from its single latent state through the affine bond formula, which is why the option-cost and replicating-portfolio machinery can reconstruct term rates cleanly without Monte Carlo noise. Calibration is a Levenberg-Marquardt least-squares fit of the mean reversion a and volatility sigma to the at-the-money cap column, with the residual measured in basis points; the smile is left to the SABR display. The rate can go negative, which is correct for a Gaussian model.

$$r(t) = f\^M(0, t) + X(t)$$

short rate = market instantaneous forward plus the mean-reverting factor; X(0) = 0

$$X\_\{k+1\} = X\_k\cdot\exp(-a\cdot dt) + \sigma\cdot\sqrt{((1 - \exp(-2a\cdot dt)) / (2a))}\cdot Z$$

exact OU transition; Z standard normal, applied with both signs across an antithetic pair

$$B(a, \tau) = (1 - \exp(-a\cdot\tau)) / a$$

the factor loading; the building block of every Hull-White bond and vol formula

$$P(t, T \mid X) = (P\^M(0,T)/P\^M(0,t))\cdot\exp(-B(\tau)\cdot X(t) - V), \quad V = (\sigma^2/4a)\cdot(1 - \exp(-2a\cdot t))\cdot B(\tau)^2$$

shifted-Gaussian bond reconstruction, τ = T - t; arbitrage-free and curve-consistent at t = 0

- **a (mean reversion).** Speed of pull-back to the forward curve, per year. Calibrated; bounded [1e-4, 0.5].
- **sigma (volatility).** Absolute (normal) volatility of the short rate, decimal per square-root year. Calibrated; bounded [1e-6, 0.05].
- **Simulation.** 500 paths (250 antithetic pairs), 30-year monthly horizon, fixed seed, multiplicative martingale correction on.

BGM / LMM (2-factor)

A LIBOR market model that evolves the forward rates directly under the spot-LIBOR measure. The instantaneous volatility of each forward is the Rebonato four-parameter parametric form, which depends only on time-to-maturity and so is time-homogeneous; two orthogonal loadings rotated by an angle beta give the two factors, and the forward-forward correlation is the cosine of the rotation difference. This compact structure is what lets a two-factor model fit the whole swaption surface.

Simulation is in Glasserman-Zhao log coordinates with optional displaced-diffusion and CEV dynamics, advanced by a predictor-corrector step that averages the drift over the start and end of the period using the same shocks. The displacement and CEV exponent tame the long-horizon upper tail, which is otherwise fuller than Hull-White's. Calibration is a Levenberg-Marquardt fit of the volatility parameters and rotation angle to the at-the-money swaption grid, accepted near or below ten basis points RMSE; because the objective is multimodal, the fit runs from two starting points and keeps the better. The displacement and CEV exponent are exogenous structural inputs, not calibrated.

$$\sigma_i(t) = volScalar \cdot [\, (a + b\cdot(\tau_i - t))\cdot \exp(-c\cdot(\tau_i - t)) + d \,]$$

Rebonato instantaneous vol of forward i; $\tau_i - t$ is time-to-maturity

$$\rho_{ij} = \cos(\, \beta\cdot(\tau_i - \tau_j)\,)$$

time-homogeneous forward-forward correlation from the rotation angle β

$$\sigma_N(ATM) \approx \sqrt{\Big(\Sigma_i \Sigma_j\, w_i\cdot w_j\cdot \hat{F}_i^{\beta_{cev}}\cdot \hat{F}_j^{\beta_{cev}}\cdot \rho_{ij}\cdot I_{ij} / T_\alpha \Big)}$$

frozen-weight swaption normal vol; $\hat{F} = F + \delta$ (displacement), w the annuity weights, I_{ij} the closed-form vol-product integral

$$d \log \hat{Y}_k = (\mu_k - ½\cdot\sigma_{loc}^2)\cdot dt + \sigma_{loc}\cdot d\tilde{W}, \quad \hat{Y}_k = F_k + \delta, \quad \sigma_{loc} = \hat{Y}^{(\beta_{cev} - 1)}\cdot\sigma$$

log-coordinate dynamics; $\beta_{cev} = 1$ collapses to displaced-lognormal

- **a, b, c, d.** Rebonato vol shape: a + b·tenor humped by exp(-c·tenor) plus a floor d. Calibrated jointly.
- **beta (rotation).** Rotation angle setting the two-factor correlation, in radians; bounded [-1, 1]. Distinct from the CEV exponent.
- **displacement and CEV exponent.** Structural tail controls (delta shift, beta_cev elasticity), chosen exogenously; the vol parameters are re-fit after they are set. The Lab default is the engine of record, displacement 0.05 and exponent 0.5: plain lognormal forwards on the monthly grid run away over the 30-year horizon, and the square-root local vol removes that at the dynamics level.

Compare

Side-by-side fan charts of the two engines at a chosen tenor, so you can see where they agree and where the BGM tail widens. The FTP and Replicating Portfolio tabs let you choose which engine drives the result; this tab is where you build intuition for that choice. Both engines pool onto the same path-and-month grid downstream, so their replicating-portfolio blends are directly comparable.

Instruments

Loans, the prepayable mortgage, the two NMD books, the deposit tractor, and their functional forms.

Fixed and floating loans

The fixed loan amortizes at a locked coupon; its cashflows are rate-path independent by construction, which is the definition of fixed. The floating loan reads the index in arrears, resets at its cadence, and re-amortizes the remaining balance over the remaining term at the new coupon, which is the practitioner-correct treatment: a rate move changes the payment, not the schedule length. Both are the clean reference cases for FTP: a fixed coupon match-funds at a fixed all-in rate, a floater funds at the reset-tenor benchmark plus TLP.

$$PMT = B_0 \cdot i \cdot (1+i)^N / ((1+i)^N - 1), \quad i = coupon / 12$$

level-pay monthly payment on notional B_0 over N months; straight-line B_0/N in the zero-rate limit

$$couponRate_m = r(t_{m-1}) + margin \quad \text{on reset months} \quad m \in \{1, 1+R, 1+2R, \ldots\}$$

floating coupon, index read in arrears at the prior month; R is the reset frequency in months

Mortgage: the calibrated competing-risks prepayment model

The Mortgage tab opens on the calibrated model; the selector at the top of the tab switches to the classroom four-factor model documented in the next subsection. The two are separate models with separate purposes, not two settings of one model. The calibrated one is a competing-risks (multi-decrement) treatment of a single-family fixed-rate cohort: two cause-specific monthly hazards, voluntary prepayment and involuntary default, running inside one survivor recursion rather than as two independent overlays. Because both decrements consume the same surviving balance, neither can be read in isolation: a month of heavy refinancing leaves less balance exposed to default, and a month of heavy default leaves less balance available to refinance. Calibration is on the public Freddie Mac Single-Family Loan-Level Dataset.

The voluntary arm is the sum of two limbs rather than a single S-curve. Turnover is the move-and-sell channel: a baseline monthly rate τ_0, ramped linearly over the first L_T months, carrying twelve fitted calendar multipliers, and attenuated by up to λ when the contract rate sits below market and the borrower is locked into a coupon they cannot replace. Refinance is a logistic in the incentive ratio cr = contract rate / market rate, with one pooled shape (steepness k, midpoint cr_0) and a ceiling that differs by credit tier, scaled by a multiplier for the calendar era being projected. There is no fitted burnout term and no global scale parameter, yet pool speed still falls through a refinancing wave: the high-ceiling tier depletes out of the shared survivor first, so burnout is emergent from the tier mix rather than a decay parameter. CPR = 1 − (1 − SMM)^12 is a reporting transform only; the arms compose in SMM space.

The default arm is a logit monthly default rate (MDR) on three arguments: an underwater flag off the mark-to-market LTV path, capped-log loan age, and credit score. The age term is ln(min(age, 84) + 1), so the hazard ramps to the 84-month knot and then plateaus, and a 29-year projection never extrapolates the age effect without bound. Severity is structural: the loss given default is read off the same mark-to-market LTV path that drives the frequency, so a house-price stress moves both at once and the correlation between them is a property of the specification instead of a judgment overlay. Defaults resolve at the event month, and recovery cash of (1 − LGD) arrives on a twelve-month lag; recoveries falling beyond the projection horizon are still discounted, never dropped.

The cohort runs as three credit-tier segments (origination FICO below 680, 680 to 759, and 760 or above) that each carry the full recursion at their own score and are aggregated at the calibration weights of roughly 11%, 41%, and 48% of loan-months. The decrement order inside a month is fixed by convention: scheduled amortization first, then default on the amortized remainder, then voluntary prepayment on what default leaves. Swapping it moves results only at the product of the two hazards. The mark-to-market LTV uses the previous month's contractual amortization factor, which is the one-month lag in the hazard.

$$SMM_j(t) = \tau_0 \cdot min(age/L_T, 1) \cdot Seas(m) \cdot [1 - \lambda \cdot lock(cr)] \; + \; e_{era} \cdot C_j/(1 + exp(-k \cdot (cr - cr_0)))$$

voluntary arm for credit tier j: turnover limb plus refinance limb, clamped to [0, 0.5]; lock(cr) = clip((1 − cr)/0.3, 0, 1), so the attenuation deepens the further the contract rate sits below market

$$MDR(t) = 1/(1 + exp(-z)), \quad z = b_0 + b_f \cdot (FICO - 745) + b_a \cdot ln(min(age, 84) + 1) + uw \cdot (b_u +$$

> $b\_fu \cdot (min(FICO, 780) - 745))$
> default arm; the 780 cap binds the interaction argument only, while the main FICO effect reads the uncapped score

> $mtmLTV(t) = LTV\_0 \cdot amort(t-1) / HPI(t), \quad uw(t) = 1 \text{ if } mtmLTV(t) > 1$
> the shared LTV path: contractual amortization against the house-price index, driving both default frequency and severity

> $LGD(t) = clip(1 - (1 - c\_liq)/mtmLTV(t), 0.02, 0.95)$
> structural severity off the same path; c_liq is the liquidation cost as a fraction of property value

> $S\_\{t+1\} = S\_t \cdot (1 - SMM\_t - MDR\_t)$
> the multi-decrement survivor, stated in composed form; the engine applies the decrements in the documented order above

- **Cohort.** Face $100M, coupon 5.5%, twelve months seasoned, 348 months remaining. Editable on the tab: remaining term, mark-to-market LTV at valuation (0.88), liquidation cost c_liq (0.25), and the flat market and discount rates (6.0% each).
- **Turnover limb.** τ_0 the fully seasoned monthly turnover before seasonality and lock-in, L_T the length of the linear age ramp, λ the deepest lock-in attenuation. The twelve calendar multipliers are fitted and are not exposed as sliders: moving one teaches nothing the limb parameters do not already show.
- **Refinance limb.** k the logistic steepness and cr_0 the midpoint, both in the incentive ratio, plus one monthly ceiling per credit tier. Raising a tier ceiling raises that tier's saturated refinancing speed and, through depletion, steepens the pool's emergent burnout.
- **Calendar era.** One multiplier on the refinance limb per era, keyed to report calendar rather than origination cohort. 2020 to 2021 is fixed at 1.0000 by construction: it is the normalization anchor that identifies the tier ceilings. The tab defaults to the 2022 to 2025 era, which is the frame the committed pins were produced under.
- **Scenario.** A parallel shock in basis points applied to the market and discount rate together, and a house-price path: base (+3%/yr) or stress (−25% over 24 months, log-linear, flat thereafter). The stress path is what pushes the cohort underwater and turns the default arm and the severity on together.
- **Where the sliders live.** Full-detail mode exposes the parameter groups and the pin card; Overview mode keeps the prose, the charts, and the comparison row. Neither mode changes the math.

Committed pins and the calibrated model's gate

The calibrated model ships with a set of committed scalars, produced by the reference engine of record and restated in the Lab so that a parameter drift shows up as a failing badge instead of a quiet number change. The Lab recomputes each quantity live from the shipped defaults and compares it to its pin. Green means the browser reproduces the engine of record to within tolerance. Red means a defect in the port, and should be read as one.

Six quantities carry pins, all read off the fixed ±200 and ±100 basis-point legs of the deterministic grid: the base weighted-average life (7.59 years), the WAL at −200bp (3.24 years) and at +200bp (9.55 years), the effective duration under a ±200bp shock (4.81 years), and the change in economic value at −200bp (+724bp) and at +200bp (−1201bp). The base present value is 97.25% of par. The asymmetry of the last pair is the negative convexity the prepayment option creates: the down-shock gain is worth roughly six-tenths of the up-shock loss, because falling rates pull the refinancing limb into its ceiling and hand the balance back before the discounting gain accrues.

Two conventions matter for reading those numbers. The change in economic value is quoted in basis points of base economic value, not of face, which is the basis the pins were struck on; a face denominator reads several percent apart on the down legs and would not gate. And the deterministic frame is a fixture: a flat market rate and a flat discount rate per scenario, shocked together, on the base house-price path and the projection era. It is not the bootstrapped curve the loan and deposit tabs price against, so a pin is a reproduction target for the engine rather than a valuation of the cohort against today's snapshot.

Moving any calibrated parameter, the calendar era, or the house-price scenario flips every pin badge to a quiet Modified chip. The numbers keep computing live, so exploration is never blocked; what is suspended is only the claim that the live number should match the pin. Reset parameters restores the gated state. The parallel-shock control is deliberately excluded from that flip, because the pinned quantities read off the fixed grid, which the shock control does not touch.

- **Tolerances.** 0.002 years on WAL and effective duration, 1.0 basis point on the change in economic value. Tight enough to catch a parameter edit, loose enough to absorb the cross-language float difference between the reference engine and the browser.
- **Pin basis.** The committed engine of record and the committed cohort conventions, as of 2026-07-18. The basis line renders in the pin card itself, so a number never appears on screen without it.

Deterministic and stochastic on the Mortgage tab

The calibrated model runs in two frames, and the tab keeps them visibly apart because they answer different questions. The deterministic frame runs the engine at a flat market and discount rate across a five-leg grid (−200, −100, 0, +100, +200 basis points, market and discount moved together), with all three credit tiers carrying their own recursion. Everything gated runs here: the pin card, the deterministic cells of the comparison row, the scenario analytics, the hazard and decrement charts, and the shock profile. Run Analytics is not required for any of it, since the grid recomputes on every control change.

The stochastic frame is the Run Analytics button, and it needs both simulations from the Rate Models section. It reads the simulated ten-year benchmark path from Hull-White and from BGM, anchors the spread so the first month matches the market rate entered on the tab, and drives the incentive ratio month by month through the engine one path at a time. The deterministic base line in those fans runs on the forward curve rather than a flat rate, so the fan chart is the only place on the tab where the term structure reaches the prepayment arm. What the bands show is dispersion: the mean CPR and surviving-balance paths with the 5th to 95th percentile envelope around them.

The path runs collapse the three credit tiers into one weighted arm instead of running each tier's recursion per path. That is a real approximation, and it is why the stochastic weighted-average life and effective duration bands are labeled as a Lab illustration and carry no pin, while the deterministic numbers above them keep the full tier structure. Treat the bands as a picture of how much rate uncertainty widens the answer, not as a second estimate of the answer.

A third comparison sits underneath both frames and is easy to confuse with them. The static benchmark is not a rate-path choice: it freezes each tier's own month-by-month SMM vector from the base run and then shocks rates, leaving the default arm live in both runs. The difference between the two is what the rate response is worth. At −200bp it reads as option cost (the value the dynamic model gives up because balances leave when they are most valuable to keep), and at +200bp as extension penalty (the value it loses because balances stay when the coupon has become unattractive). The dynamic effective duration sits below the static one for the same reason.

Mortgage: the classroom four-factor prepayment model

The second entry in the tab's model selector, kept as a teaching model: it runs on classroom parameters, carries no pin, and is where the mechanics of an S-curve prepayment overlay are easiest to see one factor at a time. Its parameter shape diagnostics redraw live as the controls move, and a constant-CPR setting is available as the rate-invariant benchmark.

A level-pay mortgage with a four-factor prepayment model. The note rate is the ten-year forward plus a fixed origination spread; prepayment is rate-path-dependent through the conditional prepayment rate (CPR), which is what gives the mortgage its embedded short option and makes its stochastic FTP differ from the single-path view. The CPR is the product of four factors: a refinancing-incentive logistic in the rate difference, a seasoning ramp, a seasonal sinusoid, and a burnout that decays with cumulative in-the-moneyness. The annualized CPR converts to a monthly single-month mortality (SMM) by the exact geometric relation, and SMM is applied to the balance net of scheduled principal.

The prepayment factors are balance-independent, which licenses a single-pass schedule. A Richard-Roll arctan refinancing variant is also implemented for comparison; it takes decimal rates and differs in the seasonality and burnout scaling, so the two are not interchangeable without rescaling.

$$rateDiff = wac - mortgageRate \cdot 100$$

refinancing incentive in percentage points; wac is the pool note rate in points, mortgageRate the current rate in decimal

$$refi = (minCpr + (maxCpr - minCpr) / (1 + \exp(-steepness\cdot(rateDiff - inflection)))) / 100$$

the refinancing-incentive S-curve, output as a decimal CPR

$$seasoning = \min(1, age/ramp), \quad seasonality = 1 + (amp/100)\cdot\sin(\pi\cdot(monthIndex - 4)/6)$$

the age ramp and the annual seasonal factor

$$burnout = \exp(-(burnoutDecay/100)\cdot\Sigma \max(0, rateDiff))$$

decays prepayment with cumulative positive in-the-moneyness

$$CPR = clip(refi \cdot seasoning \cdot seasonality \cdot burnout, 0, 1), \quad SMM = 1 - (1 - CPR)^{(1/12)}$$

the four-factor product, clamped, then converted to a monthly mortality

- **wac / steepness / inflection.** Pool note rate (6.5 points), logistic steepness (2.5), and inflection (0.5 points). The S-curve midpoint sits where the refinancing incentive is half its range.
- **minCpr / maxCpr.** Floor and ceiling annual CPR, 2% and 65%, before the seasoning, seasonality, and burnout overlays.
- **seasoningRamp / seasonalityAmp / burnoutDecay.** Months to full speed (30), seasonal amplitude (20%), and burnout decay rate (0.1).

Non-IB NMD: the decay model

The non-interest-bearing book pays no coupon and runs off on a behavioral decay model. Two limbs compose multiplicatively each month: an age-ramped baseline closure that captures relationship attrition, and a one-sided rate-driven incentive that captures depositors leaving when the market out-pays the parked balance. The incentive is a logistic in the spot rate LEVEL — the opportunity cost a 0%-paying deposit forgoes — multiplied by a regime amplifier that accelerates runoff when the rate has jumped above its trailing moving-average regime (the surge of low-yield balances now repricing out), an optional Schwartz-Torous burnout, and the balance-sensitivity multiplier g(b). The amplifier is the SSRN excursion form $1 + \alpha\cdot\max(0, r - MA)$: dormant at or below the prevailing regime, it captures the path dependency that a flat rate-level model misses.

Rate sensitivity is graded by account size through g(b), a logistic in log-balance: every dollar is rate-sensitive, but large accounts leave more opportunity cost on the table at a low deposit rate and so run off faster. The book is partitioned into editable account-size buckets, each running the same closure and incentive at its own $g(\bar{b}_i)$, and the portfolio decay is the balance-weighted aggregate. The decay surface is the outer product of the balance logistic g(b) and the rate-level logistic, the two factors visible on the tab's 3D surface. The model is explicitly illustrative; the canonical Model Sense deposit framework is separate.

$$MA_P(t) = (1/P)\cdot\Sigma_{j=1..P}\, r_{1M}(t-j), \quad excursion(t) = \max(0, r_{1M}(t) - MA_P(t))$$

trailing regime moving average and its one-sided excursion, percent per annum; the MA uses the strictly prior P months

$$base\_incentive(t) = maxGrowth + (maxDecay - maxGrowth) / (1 + \exp(-K\cdot(r_{1M}(t) - midpoint)))$$

opportunity-cost logistic on the spot rate LEVEL, percent per month; maxGrowth = 0 makes it run from 0 to maxDecay

$$g(b) = g_{min} + (g_{max} - g_{min}) / (1 + \exp(-\kappa_b\cdot(\ln b - \ln b^*)))$$

balance-sensitivity multiplier, logistic in log-balance; b* the inflection balance

$$rateIncentive_i(t) = \max(0, base\_incentive(t) \cdot regimeAmp(t) \cdot B(t) \cdot g(\bar{b}_i))$$

$regimeAmp(t) = 1 + \alpha\cdot excursion(t)$ the SSRN surge amplifier; $B(t) = \exp(-\lambda_b\cdot\Sigma_{s<t} \max(0, base\_incentive(s)))$ the burnout

$$closure(t) = closure\_steady + (closure\_initial - closure\_steady)\cdot\exp(-(t-1)/\tau)$$

age-ramped baseline closure, percent per month

$$decay_i(t) = 1 - (1 - closure(t)/100)\cdot(1 - rateIncentive_i(t)/100)$$

combined monthly decay fraction per bucket; portfolio decay is the balance-weighted aggregate

- **closure_steady / closure_initial / tau.** Long-run closure 0.75%/mo, initial 2.0%/mo, settling with a 24-month time constant. Initial equals steady disables the ramp.
- **maxRateDecay / maxRateGrowth / K / midpoint.** High-rate plateau 1.98%/mo (re-solved so the seasoned-stock WAL lands near 5 years on the 3/31 curve), growth 0 (one-sided), steepness 1.5, midpoint 1.0% rate level (well below the 3/31 operating level of ~3.7%, so the incentive is near its plateau and the level-based logistic produces a modest convexity wedge: effective duration < static discounting duration < WAL).
- **regime alpha / burnout lambda.** Regime amplification $\alpha = 2.0$ (SSRN-calibrated): the surge amplifier $1 + \alpha \cdot \max(0, r - MA_{24})$ is dormant at or below the trailing regime and amplifies in a rate spike. Burnout off by default (lambda 0).
- **g(b): g_min / g_max / b* / kappa_b.** Floor 0.15, ceiling 1.0, inflection $1.1M, steepness 1.3. Shared with the Replicating Portfolio tab, where the account-size buckets read the same shape.

IB NMD: deposit beta and the rate path

The interest-bearing book layers a dynamic deposit rate on top of the same closure decay model. The deposit rate tracks the one-month market index as a level ratio through a logistic deposit-beta S-curve, with an optional Nerlove partial adjustment that smooths the path toward target. The closure overlay then runs against the spread of market rate minus deposit rate rather than the rate-surprise spread, with the logistic midpoint shifted to re-center on the typical interest-bearing regime. The book skews toward large, rate-aware commercial balances, so it sits high on g(b) and runs off faster than the retail Non-IB book.

The default beta parameters are the in-sample MMDA estimates from Chen (2026), Dynamic Deposit Betas (SSRN 6269838), not free knobs. The Nerlove factor defaults to full pass-through; the paper estimates it near 0.47, which smooths the trajectory when set in that range. A single modeled deposit-rate path is shared across all vintages, so the FTP and margin views reuse one consistent rate.

$$\beta(r) = \beta\_min + (\beta\_max - \beta\_min) / (1 + \exp(-k \cdot (r - m)))$$

deposit-beta S-curve; r the one-month rate in percent, m the inflection level

$$D\_target(t) = \beta(r(t)) \cdot r(t)$$

level-ratio target deposit rate, percent per annum

$$D(t) = \max(0, D(t-1) + \lambda \cdot (D\_target(t) - D(t-1))), \quad D(0) = D\_target(0)$$

Nerlove partial adjustment; $\lambda = 1$ snaps to target every period

$$spread(t) = r(t) - D(t)$$

the switching incentive feeding the IB closure overlay; Non-IB uses the spot rate level directly, IB uses this spread

- **beta_min / beta_max.** Lower and upper beta asymptotes, 0.433 and 0.800 (Chen 2026, in-sample MMDA).
- **k / m.** Logistic steepness 0.566 and inflection 3.919%.
- **lambda (Nerlove).** Partial-adjustment speed in (0, 1]; default 1.0 (full pass-through), paper estimate near 0.47.
- **closure overlay.** Same form as Non-IB with the midpoint shifted to 2.0 (re-centered on the $r - D$ regime) and a higher rate-decay plateau (9.5%/mo), reflecting the commercial book's faster flight. The IB closure overlay carries no rate-level multiplier in its base form: the regime amplifier is present but off by default ($\alpha = 0$, retained for stress exploration), and the old cash-sorting salience has been dropped entirely — the beta coupon already prices in the rate level through $D = \beta(r) \cdot r$. The base rate response is the $r - D$ spread alone. The decay-incentive steepness is flattened (logisticK = 0.3): at the steeper K = 1.0 the runoff over-responded to the ±100bp shock, pushing the effective duration slightly negative on the stochastic legs (a deposit liability cannot have negative duration). Flattening it holds the rate-locked $(1 - \beta) \approx 20\%$ slice near a 2-year duration, landing a stable, positive effective duration of ~0.25 to ~0.40 years across the deterministic and stochastic scenarios, consistent with the $(1 - \beta) \cdot 2y \approx 0.40$ economic anchor; the deterministic-forward WAL eases to ~2.4 years. Under the ±100bp shock the coupon reprices at $\beta(r \pm \Delta) \cdot (r \pm \Delta)$, and the effective deposit beta including the slope term $\beta\_eff = \beta + r \cdot \beta'(r) \approx 0.79$ at the 3/31 operating rate.

Seasoned stock versus the fresh cohort

The instrument tabs model a single fresh cohort, but a real deposit book is a stack of vintages of different ages. The Lab superposes the cohort survival kernel across a steady-state age distribution to recover the seasoned stock, and reports both the stock WAL and the cohort WAL. The stock is always longer-lived than a fresh cohort, and this is a structural consequence rather than an assumption: closure declines with age, so survivors are stickier, and the survivor-weighted stock runs off more slowly than the cohort it was drawn from. On a flat rate path the stock runoff reduces to the renewal-theory identity.

$$B[m] = \Sigma\_size\ \Sigma\_age\ (sizeShare \cdot ageShare) \cdot surv\_\{size,age\}[m]$$

seasoned-stock balance as a size-by-age superposition of cohort survivals, normalized to 1

$$R(t) = \Sigma\_\{j \geq t\}\ S(j) \,/\, \Sigma\_\{j \geq 0\}\ S(j)$$

the flat-path renewal identity: remaining stock fraction at month t under constant origination

$$WAL = \Sigma\_\{m\}\ runoff\_m \cdot (m/12) + B[N] \cdot (N/12)$$

stock WAL in years, including the residual tail at the horizon to avoid a short bias

ALM Analytics

Gap views, the two FTP tabs, effective duration and economic value, and the stochastic earnings gap.

Repricing Gap and Liquidity Gap

Repricing gap buckets balances by when they reprice; liquidity gap buckets them by when principal actually runs off. Each exports to Excel with a summary plus one sheet per instrument.

Replicating Portfolio (the deposit tractor)

Represents the stable deposit balance as a ladder of moving-average pillars. A k-month pillar is k equal rolling bullets in the k-month tenor: a linear-amortizing runoff with WAL = (k+1)/2 months earning the trailing k-month average of the k-month rate. Two yields are tracked and never mixed: the steady-state carry (the trailing moving average) and the new-money yield (today's curve par). The overnight bucket is the spot rate with a half-month WAL by convention, not a one-month bullet.

The optimal blend is solved on the simulated paths as a mean-variance efficient frontier, where risk is the volatility of monthly yield changes. For the non-interest-bearing book it is a yield-space frontier (maximize yield per unit yield-volatility under a WAL cap, with the Sharpe ratio measured against the forecast-mean overnight yield). For the interest-bearing book it is a margin-space frontier that nets the modeled client rate, plus a non-negative least-squares regression of the deposit rate on the pillar yields constrained to sum to one. A Hull-White or BGM toggle selects which engine's paths drive the frontier.

This tab is an interest-rate view only: it builds the portfolio and prices its credit on term yields and repricing, with no liquidity overlay. The contingent-liquidity stress and HQLA buffer charge live on the FTP - NMD tab. The Non-IB section adds a stochastic credit fan: because the client rate is zero, the credit dispersion across paths is the structural interest-rate risk, and a shorter blend reprices faster and widens the fan.

$$WAL\_pillar(k) = (k + 1) / 2 \quad for\ k > 1, \quad 0.5 \quad for\ k \leq 1$$

weighted-average life of a k-month moving-average pillar, in months

$$\Sigma\_ij = 12 \cdot Cov(\ \Delta L\_i, \Delta L\_j\ )$$

annualized covariance of monthly pillar-yield changes; the frontier risk matrix, ridge-regularized in the solver

$$\min\ w'\Sigma w \quad s.t.\ \ \Sigma\, w\_i = 1,\ \ \Sigma\, w\_i \cdot WAL\_pillar(k\_i) \le cap,\ \ w \ge 0$$

the min-variance blend under the WAL cap; max-Sharpe is the cloud point maximizing (ret - rf)/vol

$$IB\ margin:\ ret = \Sigma\, w\_i \cdot \mu\_Y,i - \mu\_D, \quad Var(m) = w'\Sigma\_YY\, w - 2\, w \cdot \Sigma\_YD + Var(D)$$

the cross term -2 w·Σ_YD is hedge value: the min-margin-vol blend tracks the client rate

- **Corner blends.** Min-vol, max-Sharpe, max-yield (or max-margin), and Stock-WAL cap. Pick one to drive the stock-view runoff and flow-view new-money ladder.
- **Stock vs flow.** Stock view is the portfolio's expected monthly amortization (drives repricing gap and EVE). Flow view is what each month's reinvestment buys as new bullets; shorter pillars draw proportionally more new money.
- **WAL cap.** The frontier binds at the cohort-derived stock WAL (the behavioral anchor). The liquidity WAL cap is shown as a separate reference overlay, never used to fund the liquidity gap off the portfolio.

FTP - Loans

Marginal funds transfer pricing on new volume for the fixed loan, floating loan, and mortgage. Each instrument is priced as new origination match-funded against the all-in curve: par-matched against the benchmark alone (interest-rate FTP), against benchmark plus TLP (all-in FTP), and the residual is the liquidity-premium FTP. The par-match is linear in the transfer rate, so it is a direct algebraic solve with no root-finding and no optionality on a single path. The FTP margin is the locked NIM, asset yield minus all-in FTP.

The mortgage also gets a stochastic, option-adjusted view. Choose Hull-White or BGM, then run it. The Lab regenerates the mortgage cashflows along every simulated path and path-averages the par-coupon FTP; averaging the linear par-rate functional across paths is exact and preserves the covariance between discounting and cashflow timing that root-finding per path would drop. The vol value is the option premium a single forward path misses, measured as a double difference per leg. Z-spread versus OAS, both over the benchmark, cross-checks the interest-rate-leg vol value.

$$r = (\ N - \Sigma\_i\ D(t\_i) \cdot P\_i\ )\ /\ (\ \Sigma\_i\ D(t\_i) \cdot B\_\{i\text{-}1\} \cdot \Delta t\ ), \quad \Delta t = 1/12$$

par-matched transfer rate; D the discount function, P_i principal, B_{i-1} start-of-month balance

$$D\_IR(t) = P\_sofr(t), \quad D\_all\text{-}in(t) = P\_sofr(t) \cdot \exp(-TLP(t) \cdot t), \quad lpFtp = allInFtp - irFtp$$

the two discount bases; the liquidity-premium FTP is the additive residual

$$volValue\_leg = (stoch\_on - stoch\_off) - (det\_on - det\_off)$$

double difference per leg (IR, LP, all-in); 'off' forces CPR to zero, isolating the convexity a single path cannot represent

$$optionCost = Z - OAS$$

Z-spread on the deterministic prepay-on path, OAS on the per-path mean, both over the benchmark; the difference is the IR-leg option cost

FTP - NMD

Portfolio-level FTP on the standing deposit book under a constant-balance-sheet assumption, not marginal new-volume pricing. It is a snapshot of the stock FTP as of the market date. The decomposition is a matrix: columns are the book's three allocation slices (stable, non-stable, stress) and rows are the funding-credit legs (IR, TLP, HQLA buffer, all-in). The replicating weights come from the same frontier as the Replicating Portfolio tab, with the same engine toggle and corner selector.

A 30-day LCR stress runoff defines the stress slice, and the waterfall eats non-stable balances before stable. The stable slice earns its replicating-portfolio yield and TLP; the non-stable slice earns the three-month moving average of the three-month tenor and its 3M TLP; the stress slice earns the overnight rate (it is funded and held as overnight HQLA), carries no TLP, and bears the HQLA buffer charge. The buffer charge is the buffer target times the runoff times the cost of carry. The stock view uses realized trailing-MA carry; it is never interleaved with the marginal new-money view.

$$stressed = lcrRunoff, \quad nonStableSurviving = max(\, (1 - stableShare) - lcrRunoff, 0\,)$$

the runoff waterfall: stress depletes non-stable before stable

$$stableAfterStress = max(\, stableShare - max(\, lcrRunoff - (1 - stableShare), 0\,), 0\,)$$

only the spill beyond the non-stable slice reaches the stable slice

$$bufferCharge = bufferTarget \cdot lcrRunoff \cdot (carryBp / 1e4)$$

the HQLA cost-of-carry drag borne by the stress slice

$$allInCredit = irCredit + tlpCredit - bufferCharge, \quad ftpMargin = allInCredit - clientRate$$

client rate is 0 for Non-IB and the modeled D(0) for IB

- **Default assumptions.** Stable share 80%, LCR runoff 20% (Non-IB, treated as operational) or 40% (IB), buffer target 120%, cost of carry 15 bp. All are editable.
- **Stock vs marginal.** Stock legs use realized trailing-MA pillar carry; the marginal view prices the same slices at today's curve par. The two are kept on separate, labeled views.

Effective duration and economic value

Effective duration on the instrument tabs is a central finite difference of present value under a symmetric parallel shock, run as a real ALM EVE engine: only the market rate moves, while the behavioral cashflows and the discount factors are both re-derived from the shocked rates and every calibrated model parameter is held fixed. What makes it effective rather than modified is that the cashflows regenerate under each shifted path, so a genuine rate response propagates — mortgage prepayment re-fires, the deposit coupon reprices.

One design choice keeps the Non-IB regime amplifier freeze-free. The amplifier keys off the excursion r − MA: a permanent parallel shift moves r and its warmup-history MA by the same amount, leaving the excursion unchanged and the amplifier invariant. No freeze needed. The old cash-sorting salience (r/r_ref level multiplier) has been removed entirely in the current design, so there is nothing else that needs to be held at base. The EVE engine therefore re-levels the warmup history by the shock bp/100 on each shocked leg, which preserves the correct excursion without any special treatment.

Consequences differ by instrument. The Non-IB deposit pays a 0% coupon; its opportunity-cost incentive is a logistic in the spot rate LEVEL, so a +100bp shock slightly accelerates runoff and a −100bp shock slightly slows it — mild negative convexity. The decomposition hierarchy is WAL > static duration > effective duration: WAL is pure cashflow timing; static duration freezes the cashflows and shifts only the discount curve; effective duration lets both cashflows and discounting respond to the shock, compressing it further below static. The IB deposit reprices its coupon at $\beta(r)\cdot r$ and its opportunity-cost spread r − D does move under the shift, so its runoff responds and its duration sits well below WAL (the repricing franchise), staying positive because the IB decay steepness is calibrated so the runoff response does not overwhelm the discounting. The curve shock shifts the zero curve directly (DF ×exp(−Δt)); for a parallel move this matches shifting the par curve and re-bootstrapping to second order, and the par curve is what a bank actually transacts on, so the zero-shift is purely a computational simplification.

For premium or discount instruments an optional par anchor solves a single flat continuously-compounded spread on the base leg and holds it fixed across the shock legs; this is the option-adjusted-spread convention, and without it a deep-premium mortgage shows near-zero duration because the refinancing wave destroys the premium under the down-shock. The shock is applied from step 0 (the EVE convention) and discounting is on the all-in funding curve.

$$D\_eff = (PV\_- - PV\_+) / (2 \cdot PV\_0 \cdot \Delta r), \quad \Delta r = shockBp / 1e4$$

central difference; PV_0, PV_+, PV_- are path-mean present values at base and the two shocks

$$WAL = \Sigma\_c\ (principalPaid\_c / notional) \cdot (monthOffset\_c / 12)$$

principal-runoff weighted-average life in years; interest excluded, no discounting

SEG / EBP

The sensitivity-equivalent gap measures the rate-locked outstanding balance of a static-balance-sheet portfolio through the sensitivity of net interest income to a small parallel shock. It converts a plus-or-minus ten basis point NII finite difference into an equivalent dollar notional that has not yet repriced, so the outstanding SEG profile is the option-adjusted, swap-equivalent repricing schedule of the book. For vanilla instruments it reproduces the repricing-gap line exactly; for convex instruments it bakes the linear swap-equivalent of the embedded option into the gap. The equivalent balance profile is the companion existing-book balance trajectory, with deterministic and Monte Carlo percentile fans.

Underneath sits a static-balance-sheet assumption: each month the existing book's runoff is replaced by a fresh vintage struck at the current shocked-rate environment, so total balance holds at notional. The carrying spread of each new vintage is referenced off the unshocked base curve on purpose, so the parallel shock propagates fully into the new coupon rather than being absorbed. The first month's coupon is locked before the shock (the NII convention), distinct from the EVE shock that moves step 0.

$$CumulativeSEG(t) = (NII\_up(t) - NII\_down(t)) \cdot 12 / \Delta r, \quad \Delta r = 2 \cdot SHOCK\_BP/1e4 = 0.0020$$

the central NII finite difference, annualized; SHOCK_BP = 10

$$OutstandingSEG(t) = side\_sign \cdot (notional - CumulativeSEG(t))$$

the rate-locked outstanding balance; side_sign is +1 for assets, -1 for liabilities

Glossary

All-in FTP — The transfer rate par-matched against the benchmark curve plus the term liquidity premium. The full funding cost the desk pays.

Antithetic pairing — A variance-reduction scheme that draws one shock per pair of paths and applies it with both signs, halving the variance of quantities roughly linear in the shock.

BGM / LMM — The LIBOR market model: a multi-factor forward-rate simulation calibrated to the swaption surface. One of the Lab's two rate engines.

Burnout — The decay of prepayment or deposit rate-sensitivity with cumulative in-the-moneyness, modeled as exp(-rate · cumulative incentive). Off by default for the deposit books.

CEV / displacement — The constant-elasticity-of-variance exponent and the additive shift applied to forwards in the BGM dynamics; structural controls on the upper tail, set exogenously.

Closure — The age-ramped baseline attrition of a deposit cohort, in percent per month, independent of rates; the floor under the rate-driven decay.

Committed pin — A scalar produced by the reference engine of record and restated in the Lab, recomputed live and badge-gated so a parameter drift shows as a failure rather than a quiet number change. Departing the calibrated configuration suspends the pin and shows a Modified chip.

Competing risks — Two cause-specific monthly hazards, voluntary prepayment and involuntary default, consuming the same surviving balance inside one multi-decrement recursion rather than running as independent overlays.

CPR / SMM — Conditional prepayment rate (annualized) and single-month mortality; SMM = 1 - (1 - CPR)^(1/12).

Deposit beta — The fraction of a market-rate move that passes through to the deposit rate, modeled as a logistic S-curve beta(r); the IB deposit rate is beta(r)·r.

Deposit tractor — The replicating portfolio built from moving-average pillars; a k-month pillar is k rolling bullets in the k-month tenor, WAL (k+1)/2 months.

EVE — Economic value of equity: the present value of assets minus liabilities, sensitive to the rate-shock applied. The Lab shocks from step 0 for EVE.

Effective duration — A central finite difference of PV under a permanent parallel shock, run as an EVE engine: cashflows and discount factors regenerate from the shocked rates while calibrated parameters are held fixed. Genuine rate response propagates — coupon repricing, prepayment, and the Non-IB opportunity-cost runoff (which shortens slightly as rates rise, a mild negative convexity with the operating level on the saturated plateau). The regime amplifier keys off the excursion r − MA, invariant to a parallel shift, so it needs no freeze.

FTP margin — The locked net interest margin: asset yield minus all-in FTP for assets, or all-in FTP credit minus deposit cost for liabilities.

g(b) balance sensitivity — The logistic-in-log-balance multiplier g_min + (g_max - g_min)/(1 + exp(-kappa_b·(ln b - ln b*))) that grades deposit rate-sensitivity by account size.

HQLA buffer charge — The cost-of-carry drag on the stress slice of the NMD FTP: buffer target times stressed runoff times the cost of carry.

Hull-White — A one-factor short-rate model with an exact OU transition and a martingale correction. One of the Lab's two rate engines.

IR FTP — The interest-rate component of FTP, par-matched against the benchmark curve alone, before any liquidity premium.

LCR runoff — The 30-day stressed outflow assumption from the liquidity coverage ratio, used to carve the contingent-liquidity stress slice out of the NMD book.

LP FTP — The liquidity-premium component of FTP: all-in FTP minus IR FTP. Additive by construction.

Martingale correction — A multiplicative per-step rescale of simulated discount factors so their path-mean equals the market discount factor at every horizon, removing discretization drift.

Nerlove partial adjustment — A geometric-lag smoothing of the deposit rate toward its target, D(t) = D(t-1) + lambda·(target - D(t-1)); lambda = 1 is full pass-through.

MDR / CDR — Monthly default rate and its annualized form, CDR = 1 - (1 - MDR)^12. The involuntary arm of the mortgage competing-risks model, alongside SMM on the voluntary side.

NMD — Non-maturity deposit: checking, savings, money-market balances with no contractual maturity. Modeled behaviorally.

OAS — Option-adjusted spread: the flat spread over the simulated discount paths that prices an instrument to par, capturing the embedded option.

Par-match — Solving the transfer rate that prices a cashflow stream to par on a given discount curve; linear in the rate, so a direct algebraic solve.

Pillar — A k-month moving-average building block of the replicating portfolio: k equal rolling bullets in the k-month tenor.

Rebonato volatility — The four-parameter (a, b, c, d) parametric instantaneous-volatility form of BGM, with a rotation angle beta setting the two-factor correlation.

Replicating portfolio — A blend of moving-average pillars whose cashflow and yield profile stands in for the behavioral deposit balance.

SABR — The stochastic-volatility smile model; here the Bachelier (beta = 0) normal-vol expansion used to display the volatility surface.

Regime amplifier — The Non-IB surge multiplier on the rate-incentive limb, $1 + \alpha \cdot \max(0, r - MA)$: when the spot rate exceeds its trailing moving-average regime, the low-yield deposits accumulated in the prior low-rate era reprice out faster. One-sided and dormant at or below the regime; the SSRN excursion form (Chen). It replaced the earlier rate-level cash-sorting multiplier and, keyed off the excursion, is invariant to a parallel EVE shock so it needs no freeze. The IB book uses no such multiplier — its beta coupon already prices in the rate level.

Seasoned stock vs cohort — The age-superposed deposit book versus a single fresh cohort; the stock is longer-lived because closure declines with age and survivors are stickier.

SEG / EBP — Sensitivity-equivalent gap and equivalent balance profile: the rate-locked outstanding balance from the NII sensitivity, and the existing-book balance trajectory.

Structural LGD — Loss given default read off the same mark-to-market LTV path that drives default frequency, clip(1 - (1 - c_liq)/mtmLTV, 0.02, 0.95), so a house-price stress moves severity and frequency together.

TLP — Term liquidity premium: the spread of the institution's term-funding curve over the matching-tenor benchmark (e.g. FHLB advances at a U.S. bank). Zero at the overnight bucket by convention.

Vol value — The option premium a single deterministic rate path misses, measured as the stochastic minus deterministic FTP increment.

WAL — Weighted-average life: the principal-weighted average time to repayment of a cashflow stream.

Z-spread — The static spread over the deterministic discount curve that prices a cashflow stream to par.